

Project

Category: Classification

Title: Predicting Sleep Disturbance Risk from Gaming Behavior Using Machine Learning

Dataset-1: Gaming and Mental Health Dataset - Kaggle

<https://www.kaggle.com/datasets/shaistashahid/gaming-and-mental-health>

This dataset examines gaming behavior and its potential impact on sleep patterns and daily lifestyle. It includes demographic details, gaming duration, preferred genres, and sleep-related indicators. The dataset is designed to help researchers, students, and data analysts explore patterns between excessive gaming and lifestyle factors such as sleep quality and disruption.

Dataset-2: A Multidimensional Dataset of Student Gaming Behavior and Academic Correlates in Bangladesh

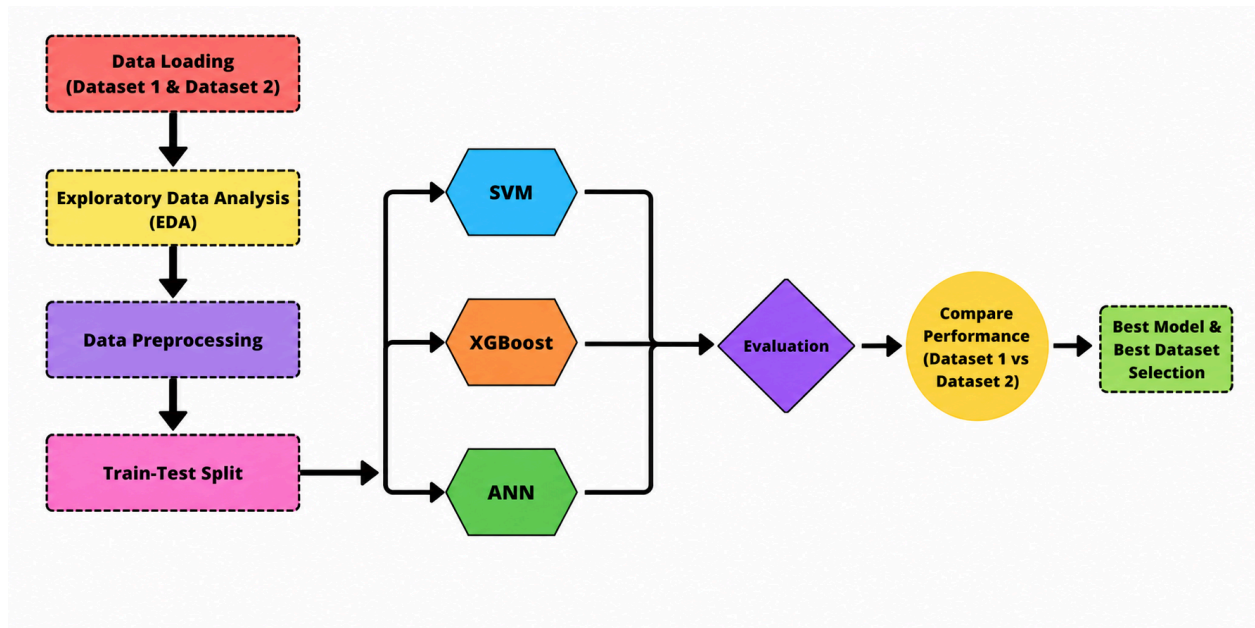
<https://zenodo.org/records/15633261>

<https://doi.org/10.1038/s41597-025-05614-4> (Paper)

This dataset contains anonymized survey responses from 1,061 Bangladeshi university students collected in March–April 2023. It includes data on gaming habits (platform, daily play time, age started, emotional responses), academic performance (CGPA, exam results, attendance), lifestyle and demographics (age, gender, sleep, social time) and emotional/physical effects of gaming (stress relief, fatigue, inability to play).

Abstract: This project aims to analyze the relationship between gaming behavior and sleep health using supervised machine learning techniques. The objective is to predict whether an individual's gaming habits are associated with poor sleep quality or increased sleep disturbance risk. Two separate datasets related to gaming behavior, mental health and sleep patterns are analyzed independently to evaluate predictive performance across different data sources. Behavioral, environmental and demographic factors are examined to identify the variables that contribute most significantly to sleep-related issues. Multiple classification models, including Support Vector Machine (SVM), XGBoost and Artificial Neural Network (ANN) are implemented and compared to determine the most reliable approach for predicting sleep disturbance risk.

Methodology



Literature Review

Paper-1 (Plaban): <https://doi.org/10.1192/j.eurpsy.2022.2113>

Description: The paper “The relationship between sleep problems and gaming addiction in adults” investigates how excessive gaming behavior affects sleep quality in young adults. The researchers conducted a survey study with 109 participants aged 18-40. They measured gaming addiction using the Gaming Addiction Scale (GAS) and evaluated sleep quality using the Pittsburgh Sleep Quality Index (PSQI). The results showed a clear connection between gaming addiction and sleep problems. About 25.7% of participants were classified as addicted gamers, and 59.6% experienced poor sleep quality. Higher gaming addiction scores were strongly linked to poorer sleep quality, more sleep disturbances, and even increased use of sleep medication. The study concludes that heavy gaming habits can negatively affect sleep patterns and overall daily functioning.

This paper is relevant to the project because it demonstrates a clear relationship between gaming behavior and sleep problems. The study found that individuals with higher gaming addiction scores were more likely to experience poor sleep quality and frequent sleep disturbances. This supports the core idea of the project, which is to use machine learning to predict sleep disturbance risk based on gaming habits and demographic factors. By confirming that gaming behavior can significantly influence sleep health, the paper provides strong academic support for building predictive models that analyze gaming patterns to identify individuals who may be at risk of sleep-related issues.

Paper-2 (Plaban): <https://pubmed.ncbi.nlm.nih.gov/35734653>

Description: The paper “Prevalence of Gaming Addiction and its Impact on Sleep Quality: A Cross-Sectional Study from Pakistan” examines how common gaming addiction is and how it affects sleep quality. The researchers conducted an online survey with 618 adults using the Gaming Addiction Scale (GAS) to measure gaming addiction and the Pittsburgh Sleep Quality Index (PSQI) to evaluate sleep quality. The results showed that 57% of participants played online games, and about 12.5% of gamers were classified as addicted. The study found that individuals with gaming addiction had poorer sleep quality, shorter sleep duration, more sleep disturbances, and greater daytime dysfunction compared to non-addicted players. Overall, the research concludes that gaming addiction is significantly associated with poor sleep health.

This paper is relevant to the project because it provides strong evidence that gaming addiction is closely linked to sleep problems. The study shows that factors related to gaming behavior can significantly influence sleep quality, which directly supports the goal of the project which is to

predict sleep disturbance risk based on gaming habits using machine learning. The findings help justify the use of gaming-related features (such as gaming time and behavioral patterns) in predictive models and demonstrate that analyzing these variables can help identify individuals who may be at risk of sleep disturbances.

Paper-3 (Plaban):

https://www.researchgate.net/profile/Tanveer-Hasan-9/publication/381654411_Prediction_of_Bangladeshi_Urban_Children's_Mental_Health_for_the_Effect_of_Mobile_Gaming_Using_Machine_Learning/links/66792ab61dec0c3c6f9fa877/Prediction-of-Bangladeshi-Urban-Childrens-Mental-Health-for-the-Effect-of-Mobile-Gaming-Using-Machine-Learning.pdf

Description: The paper “Prediction of Bangladeshi Urban Children’s Mental Health for the Effect of Mobile Gaming Using Machine Learning” focuses on how excessive mobile gaming can affect the mental health of children in urban areas of Bangladesh. The researchers collected data from 1196 children aged 5-17 through surveys completed by their parents. Using this data, they applied several machine learning algorithms which are KNN, Support Vector Machine (SVM), Naive Bayes, and Random Forest to predict the level of mental health impact caused by gaming. The results categorized children into three groups: Serious, Partial and Normal mental health impact. Among the tested models, SVM performed the best with an accuracy of 92.75%, showing that machine learning can effectively predict mental health risks related to gaming behavior.

This paper is relevant because it also uses machine learning to analyze the impact of gaming behavior on health-related outcomes. While the study focuses on children’s mental health, the core idea is similar to the project, which aims to predict sleep disturbance risk from gaming behavior. The paper shows how behavioral data related to gaming can be used to train classification models and make predictions about health conditions. It also demonstrates the effectiveness of algorithms like SVM, which supports the approach of comparing multiple machine learning models to find the most accurate predictor in gaming-related health studies.

Paper-4 (Plaban): <https://link.springer.com/article/10.1186/s44167-024-00045-9>

Description: The paper “Machine Learning in Physical Activity, Sedentary, and Sleep Behavior Research” explains how machine learning can be used to analyze complex health-related behavior data such as physical activity, sedentary habits, and sleep patterns. The authors highlight that these behaviors are interconnected and generate large amounts of data (especially from wearable devices), which traditional statistical methods often struggle to analyze effectively.

To address this, the paper describes how both supervised and unsupervised machine learning models can identify patterns, classify behaviors and predict health outcomes from complex datasets. It also outlines the typical machine learning pipeline, including data exploration, feature engineering, model development, and model evaluation. Additionally, the paper discusses commonly used algorithms such as decision trees, random forests, and support vector machines and explains how they can be applied to tasks like behavior classification, pattern discovery and predicting health-related outcomes related to sleep and activity.

This paper is relevant to my project because it explains how machine learning can be used to analyze complex behavioral data and predict health outcomes such as sleep quality. It highlights that lifestyle behaviors and daily habits can influence sleep patterns, which supports the idea that gaming behavior may contribute to sleep disturbances. The paper also discusses the use of supervised machine learning models and evaluation metrics for classification tasks, which aligns with my project's goal of predicting sleep disturbance risk using features like gaming hours, platform, and demographic factors. Overall, it provides methodological support for applying machine learning models to study the relationship between behavior and sleep health.

Paper 5 (Minhaz): <http://arno.uvt.nl/show.cgi?fid=170844>

Description: This study utilized smartphone application usage data (including gaming, social media, and video streaming) to predict sleep quality using machine learning. It compared Multiclass Logistic Regression, Random Forest, XGBoost, and Multilayer Perceptron to classify sleep efficiency and duration based on behavioral digital logs.

This research is highly analogous to your Kaggle project as it directly uses digital behavior categories to forecast sleep outcomes. It highlights how app-specific data, such as daily gaming hours, can serve as a robust predictor feature. Furthermore, their methodology of comparing tree-based models (Random Forest, XGBoost) against neural networks offers a solid baseline for your own algorithm evaluation when determining the most reliable predictive model.

Paper-6 (Minhaz): <https://ieeexplore.ieee.org/document/11380705/>

Description: Researchers applied a machine learning framework on a dataset of university students to assess the impact of gaming habits on psychological wellbeing and academic risk. Algorithms such as Logistic Regression, Random Forest, and K-Means Clustering were deployed to identify risk patterns based on gaming intensity and platform type.

This study provides a strong rationale for your predictor features, specifically targeting your `daily_gaming_hours` and `gaming_platform` variables. By demonstrating that the gaming environment and intensity significantly impact overall wellbeing (which inherently encompasses sleep health), it validates your feature selection. Their use of clustering to categorize heavy versus recreational gamers could also inspire a valuable feature engineering step in your own pipeline before classification.

Paper-7 (Minhaz):

https://www.researchgate.net/publication/380275509_EVALUATING_MACHINE_LEARNING_MODELS_FOR_PREDICTING_SLEEP_DISORDERS_IN_A_LIFESTYLE_AND_HEALTH_DATA_CONTEXT

Description: This paper evaluated multiple machine learning models (Random Forest, XGBClassifier, Support Vector Machines, and Decision Trees) to predict sleep disorder categories (None, Insomnia, Sleep Apnea) based on comprehensive lifestyle and health data, including screen time, stress, and physical activity.

This study serves as an excellent architectural blueprint for your project's modeling phase. It tackles the exact challenge of multi-class sleep disorder prediction using behavioral lifestyle inputs. Their comparative analysis of SVMs and ensemble methods (like Random Forest) will help you structure your own model evaluation and justify your final choice of the most reliable predictive algorithm for the Kaggle dataset.

Paper-8 (Yeamin): <https://link.springer.com/article/10.1186/s12889-025-21880-8>

Description: This study investigates the relationship between PUBG gaming behavior and health outcomes such as sleep disorders, migraine symptoms, and quality of life using a machine learning approach. Data were collected from 500 PUBG players and categorized based on gaming intensity levels (low, medium, and high). The research applied a Multilayer Perceptron (MLP) neural network along with statistical analysis to examine how gaming addiction, gaming duration, and behavioral factors influence sleep quality and related health outcomes.

This research is closely aligned with the objective of predicting sleep disturbance risk from gaming behavior using machine learning techniques. It demonstrates that gaming intensity and behavioral gaming patterns can significantly influence sleep-related outcomes. The use of a neural network model to analyze gaming behavior and sleep disturbances also provides a methodological reference for applying machine learning algorithms to classify sleep quality and evaluate the predictive performance of different models.

Paper-9 (Yeamin):

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10557018/>

Description:

This study developed a machine-learning–based model to predict the risk of sleep disorders, including insomnia, obstructive sleep apnea (OSA), and comorbid insomnia and sleep apnea (COMISA). The research used questionnaire responses and demographic information from clinical datasets to build predictive models. The study primarily applied the Extreme Gradient Boosting (XGBoost) algorithm and evaluated feature importance using SHAP analysis. The model achieved strong predictive performance with an AUROC greater than 0.897, showing that machine learning can accurately identify individuals at risk of sleep disorders using lifestyle and health-related variables.

This research supports the objective of predicting sleep disturbance risk from behavioral patterns by demonstrating that machine learning models can effectively identify sleep-related health risks using lifestyle and demographic variables. The use of advanced classification models highlights the potential of machine learning techniques for detecting sleep disturbances and improving early screening and diagnosis.

Paper-10 (Yeamin):

<https://www.researchgate.net/publication/380275509>

Description:

This study evaluated the effectiveness of several machine learning models in predicting sleep disorders using lifestyle and health-related data. The dataset contained 400 individual records

with features including demographic information, sleep metrics, and lifestyle factors that influence sleep quality. Multiple machine learning algorithms, including Decision Tree, Random Forest, Support Vector Machine (SVM), and Logistic Regression, were implemented and compared to determine the most accurate model for sleep disorder classification.

The study demonstrates that lifestyle behaviors and demographic characteristics can be effectively used by machine learning algorithms to classify sleep disorders. The comparison of multiple models highlights the importance of evaluating different machine learning approaches when developing predictive systems for sleep disturbance detection.

Codebase

Google Colab:  10_12_22101831_24241197_23241007.ipynb

GitHub: <https://github.com/Plaban-8/Gaming-Behavior-Sleep-Analysis>

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